

Statistics II: Introduction to Inference

Problem set 3

1. Let F_θ be **exponential**(λ) distribution with the pdf

$$f_\lambda(x) = \lambda \exp\{-\lambda x\}, \quad \lambda > 0, x > 0,$$

and $\theta = \lambda$. Find the UMVUE of the population mean $\psi(\theta) = \theta^{-1}$. Find the MSE of the UMVUE.

2. Let F_θ be **Poisson**(λ) distribution and $\theta = \lambda$. Find the UMVUE of θ . Find the MSE of the UMVUE.
3. Let F_θ be **normal**($\mu, 1$) and $\theta = \mu$. Consider the class of linear estimators $T_1(\mathbf{X})$ for μ . Show that minimizing the MSE of $T_1(\mathbf{X})$ with respect to $\mathbf{l}$ does not lead to a valid estimator.
4. Suppose $n = 10$ individuals throw a biased die independently, continuing until they roll a *six*. The number of tosses for each person are as follows:

15 12 9 34 5 1 15 44 16 5.

We are interested in estimating the probability of rolling a *six*, denoted as p . Find the UMVUE of p .

5. Let $W_1, \dots, W_k$ be unbiased estimators of a parameter θ with known variances $\text{var}(W_i) = \sigma_i^2$, $i = 1, \dots, k$. Find the best unbiased estimator of θ of the form $\sum_{i=1}^k a_i W_i$.
6. Suppose that when the radius of a circle is measured, a random error is made, which is modeled as $N(0, \sigma^2)$. If n repeated independent measurements are made, then find an unbiased estimator of area of the circle. Is it the UMVUE?
7. Let the random variable X take on values $-1, 0, 1, \dots$ with probabilities

$$P(X = -1) = \theta, \quad P(X = k) = (1 - \theta)^2 \theta^k; \quad k = 0, 1, \dots; \quad 0 < \theta < 1.$$

Show that $U = U(X)$ is an unbiased estimator of θ if

$$U(k) = -kU(-1); \quad k = 0, 1, \dots$$

or equivalently if

$$U(k) = ak, \quad \text{for all } k = -1, 0, 1, \dots$$

and some a .

8. Let $X_1, \dots, X_n$ be a random sample from **Bernoulli**(θ), $\theta \in (0, 1)$. Show that

$$\delta = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

depends only on the complete sufficient statistics and hence find UMVUE of $\theta(1 - \theta)$.

9. Let $X_1, \dots, X_n$ be a random sample from **Uniform**($0, \theta$), find UMVUE of θ^k for any integer $k > -n$.
10. Let $X_1, \dots, X_n$ be a random sample from

$$f(x) = \begin{cases} e^{-(x-\alpha)}, & x > \alpha, \\ 0, & \text{otherwise.} \end{cases}$$

It can be shown that $X_{(1)}$ is a complete and sufficient statistic for α . Find the UMVUE of α .